

Chapter 6

Simple Pricing

Learning Objectives

- 6.1 Define individual demand and aggregate (or market) demand.
- 6.2 Describe pricing and the steps to arrive at an optimal price.
- 6.3 Estimate the price elasticity of demand using the appropriate formula and explain the factors affecting elasticity.
- 6.4 Explain how to use different elasticities including income elasticity, cross-price elasticity, and advertising elasticity to forecast changes in demand.
- 6.5 Use stay-even analysis to determine the quantity change required to offset a price change.

Chapter 6 – Main Points

- **Aggregate demand** or market demand is the total number of units that will be purchased by a group of consumers at a given price.
- Pricing is an extent decision. Reduce price (increase quantity) if $MR > MC$. Increase price (reduce quantity) if $MR < MC$. The optimal price is where $MR = MC$.
- **Price elasticity of demand**, $e = (\% \text{ change in quantity demanded}) \div (\% \text{ change in price})$
- Estimated price elasticity $= [(Q_1 - Q_2)/(Q_1 + Q_2)] \div [(P_1 - P_2)/(P_1 + P_2)]$ is used to estimate demand from a price and quantity change.
- If $|e| > 1$, demand is **elastic**; if $|e| < 1$, demand is **inelastic**.
- $MR > MC$ implies that $(P - MC)/P > 1/|e|$; that is, the more elastic demand is, the lower the price
- Four factors make demand more elastic:
 - Products with close **substitutes** (or distant **complements**) have more elastic demand.
 - Demand for brands is more elastic than industry demand.
 - In the long run, demand becomes more elastic.
 - As price increases, demand becomes more elastic.
- **Income elasticity**, **cross-price elasticity**, and advertising elasticity are measures of how changes in these other factors affect demand.
- It is possible to use elasticity to forecast changes in demand: $\% \Delta \text{Quantity} \approx (\text{factor elasticity}) * (\% \Delta \text{Factor})$.
- **Stay-even analysis** can be used to determine the quantity change required to offset a price change. The stay-even quantity is $\% \Delta Q = \% \Delta P / (\% \Delta P + \text{margin})$

Warm-Up

- Think of something you buy regularly: coffee, snacks, streaming, etc. What would happen if the price went up by 10%? Would you still buy it? Would you switch to something else?

Chapter 6 - Simple Pricing and Marginal Analysis

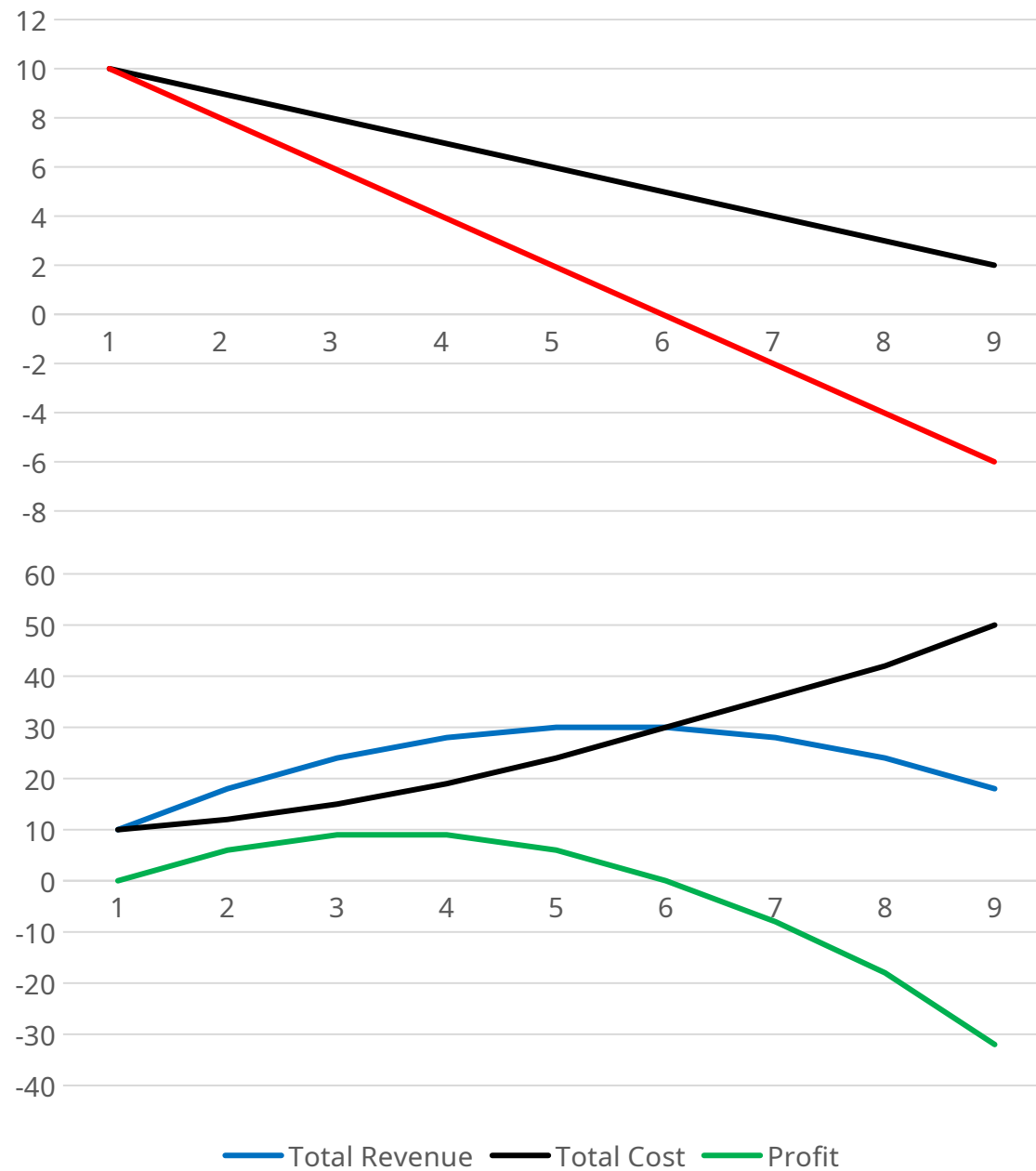
- Demand curves provide a crucial insight into pricing and marginal analysis
- When the price of a good decreases, more of the good is sold
- When the price of a good increases, less of the good is sold
- This may incorrectly lead us to believe that in order to gain more in profit we need to lower price to gain more sells
- A much simpler way, and as theory tells us, is to raise price and sell less.
- While this may seem counterintuitive, the marginal analysis behind it shows it is actually a good way to increase profit.

A Simple Analysis

Consider the following table:

Price	Quantity	Total Revenue	Marginal Revenue	Total Cost	Marginal Cost	Profit
10	1	10	10	10	10	0
9	2	18	8	12	2	6
8	3	24	6	15	3	9
7	4	28	4	19	4	9
6	5	30	2	24	5	6
5	6	30	0	30	6	0
4	7	28	-2	36	6	-8
3	8	24	-4	42	7	-18
2	9	18	-6	50	8	-32

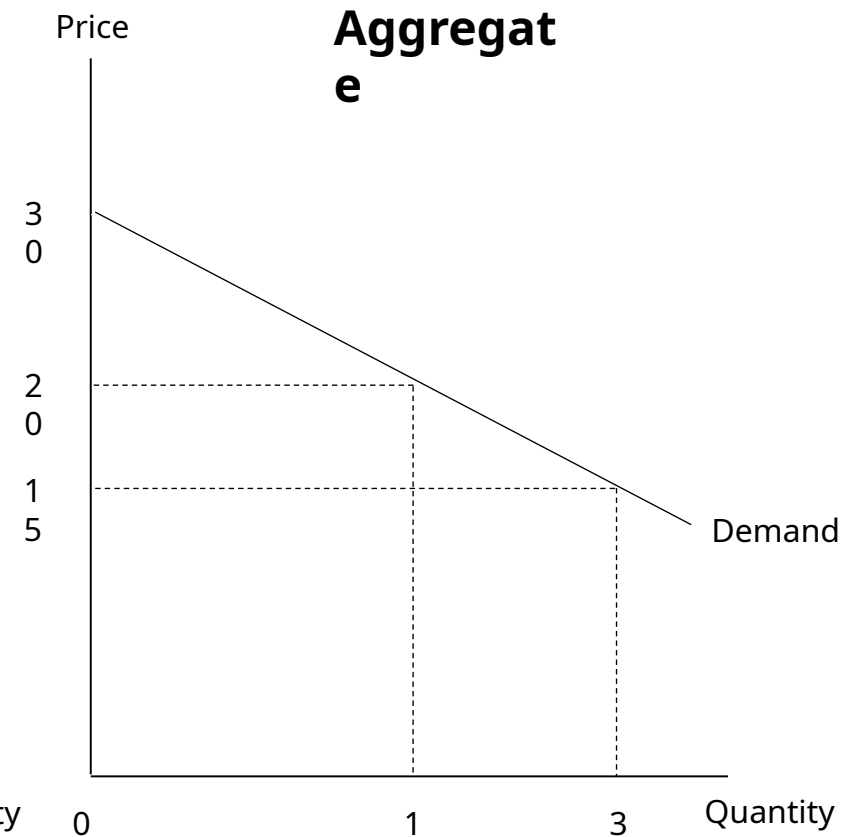
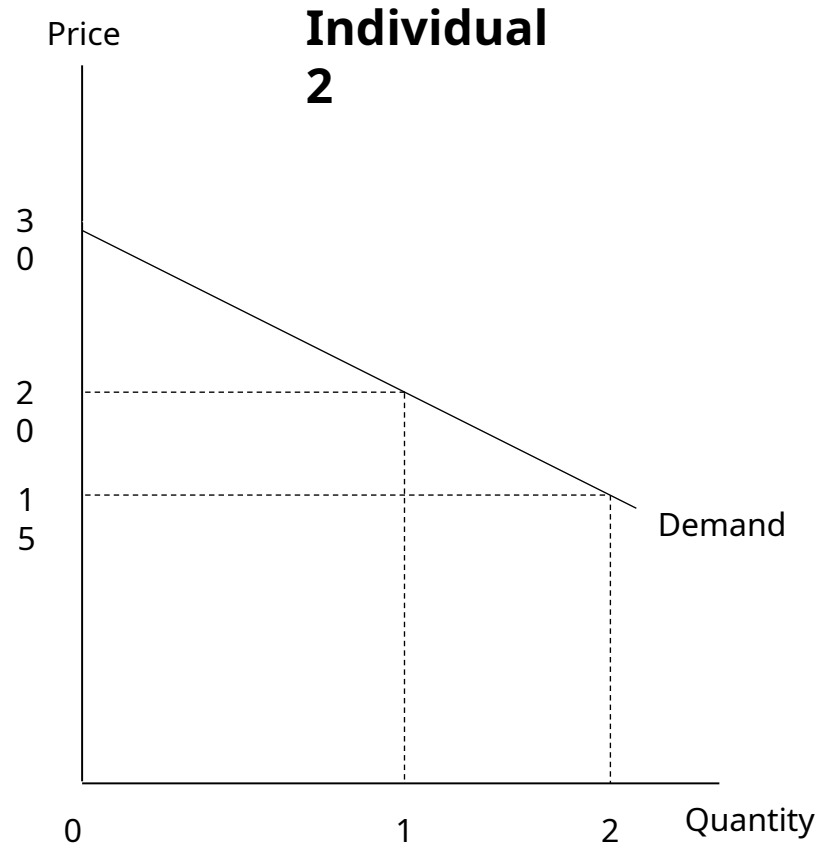
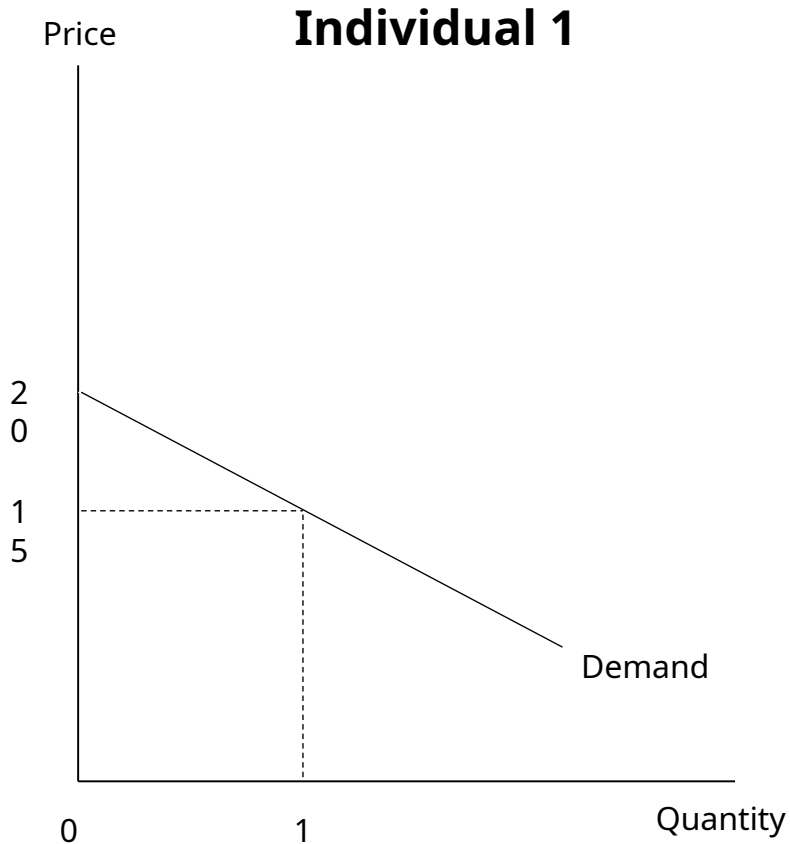
Demand



Individual vs Aggregate Demand

- The previous table and graph could represent individual or aggregate demand.
- What is the difference?
- Individual demand is based on preferences and income from the individual and only measures what they are willing and able to pay for a good.
- Aggregate demand is the horizontal summation of all individual demand curves in a particular market

Individual vs Aggregate Demand



Discussion Activity 1

- If you lower the price of a product, how do individual vs. aggregate demand respond? Why is this distinction important?
- Think about how many people are affected and whether their behaviors are independent or not.

Elasticity of Demand

- Determining marginal revenue and marginal cost are essential for us to know the optimal price
- We use the price elasticity of demand to help us determine the optimal price as well.
- **Price elasticity of demand** (P.E.D.) – the percentage change in quantity over the percentage change in price
- The P.E.D. is easily obtained from changing your price and observing the change on sales
- The simplest elasticity (measure of response) of demand is found by using the following:

$$E_D = \%Q/\%P$$

- A 5% increase in price leads to a 10% decrease in quantity demanded
 - The elasticity of demand is $-10\%/5\% = -2$

Elasticity Formulas

- Two useful price elasticity of demand formulas:

- Point Price Elasticity

$$\epsilon_D = \frac{\partial Q}{\partial P} * \frac{P}{Q}$$

- Arc Price Elasticity

$$E_D = \frac{Q_2 - Q_1}{P_2 - P_1} * \frac{P_2 + P_1}{Q_2 + Q_1}$$

Point Price Elasticity Example (not in quizzes or tests)

- Evaluate the point price elasticity of demand when quantity is 40 and price is \$30.

$$Q = 100 - 2P$$

$$\varepsilon_D = \frac{\partial Q}{\partial P} * \frac{P}{Q} \qquad \varepsilon_D = -2 * \frac{30}{40} = -1.5$$

- A 1% increase in price leads to a 1.5% decrease in quantity demanded

Arc Price Elasticity Example

- BestBuy lowered the price of its in-house brand TVs. Originally, the TV sold for \$500, and BestBuy sold 5,000 a month. BestBuy lowered the price to \$450 (a 10% change) and sales increased by 250 a month (a 5% change). What is the arc price elasticity of demand for these TVs?

$$\frac{5250 - 5000}{450 - 500} \times \frac{450 + 500}{5250 + 5000} =$$

$$\frac{250}{-50} \times \frac{950}{10250} = -\frac{19}{41}$$

$$= -0.463$$

- A 1% increase in price leads to a 0.463% decrease in quantity demanded.

Activity – What's the Elasticity?

Scenario A:

- A company raises the price of its wireless earbuds from \$60 to \$75.
- As a result, quantity sold falls from 2,000 units to 1,500 units.

Scenario B:

- A coffee shop lowers its price from \$5.00 to \$4.50, and daily sales rise from 100 to 130 cups.

Questions:

1. Use the arc elasticity formula to calculate the price elasticity of demand for each scenario.
2. Is demand elastic, inelastic, or unit elastic in each case?
3. What pricing advice would you give each company based on your result?

Elasticity Classifications

- $ED > |1|$: demand is **elastic**. A 1% increase (decrease) in price leads to a more than 1% decrease (increase) in quantity demanded
- $ED < |1|$: demand is **inelastic**. A 1% increase (decrease) in price leads to a less than 1% decrease (increase) in quantity demanded
- $ED = |1|$: demand is **unit elastic**. A 1% increase (decrease) in price leads to a 1% decrease (increase) in quantity demanded
- $ED = 0$: demand is **perfectly inelastic**. A 1% increase (decrease) in price leads to no change in quantity demanded
- $ED = \infty$: demand is **perfectly elastic**. A 1% increase (decrease) in price leads to quantity demanded falling to 0.

Activity – What's the Elasticity?

Scenario A: Wireless Earbuds

- $P1 = \$60$, $P2 = \$75$
- $Q1 = 2,000$, $Q2 = 1,500$
- Arc Price Elasticity:
 - $((1,500 - 2,000) / (75 - 60)) \times ((75 + 60) / (1,500 + 2,000)) = -1.29$
- **Interpretation:** Demand is elastic ($|E| > 1$). Price increase leads to a more than proportional drop in quantity. Consider lowering the price total revenue is likely falling (we'll get to that in a bit).

Activity – What's the Elasticity?

Scenario B: Coffee Cups

- $P1 = \$5.00$, $P2 = \$4.50$
- $Q1 = 100$, $Q2 = 130$
- Arc Price Elasticity:
$$\left(\frac{130 - 100}{4.50 - 5.00} \right) \times \left(\frac{4.50 + 5.00}{130 + 100} \right) = -2.48$$
- **Interpretation:** Demand is very elastic ($|E| > 1$). A small price drop significantly boosts quantity sold. The price cut may have increased total revenue, continue promoting volume.

Discussion Activity 2

- Why might the demand for luxury handbags be more elastic than the demand for insulin?
- Consider necessity, availability of substitutes, and budget share.

Activity – Elastic or Inelastic?

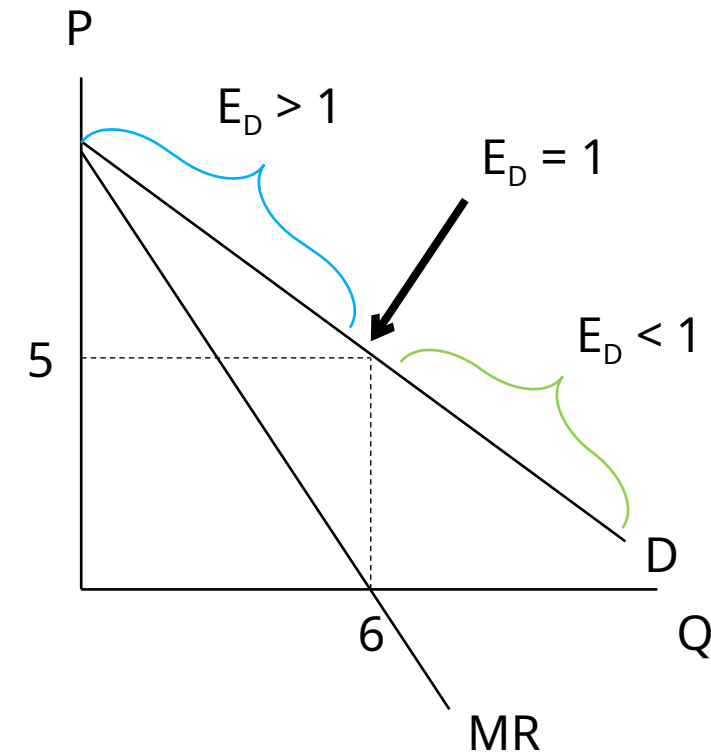
Product	Elastic or Inelastic?	Why?
Gasoline (short run)		
Designer sunglasses		
Epipen		
Streaming subscriptions		
Brand-name cereal		

Activity – Elastic or Inelastic?

Product	Elastic or Inelastic?	Why?
Gasoline (short run)	Inelastic	Few substitutes, necessity for commuting, hard to adjust usage quickly
Designer sunglasses	Elastic	Luxury item, many substitutes, price-sensitive customers
Epipen	Inelastic	Life-saving drug, no close substitutes, urgent need
Streaming subscriptions	Elastic	Many alternatives, discretionary spending, low switching costs
Brand-name cereal	Elastic	Substitutes exist (generic/store brands), not a necessity

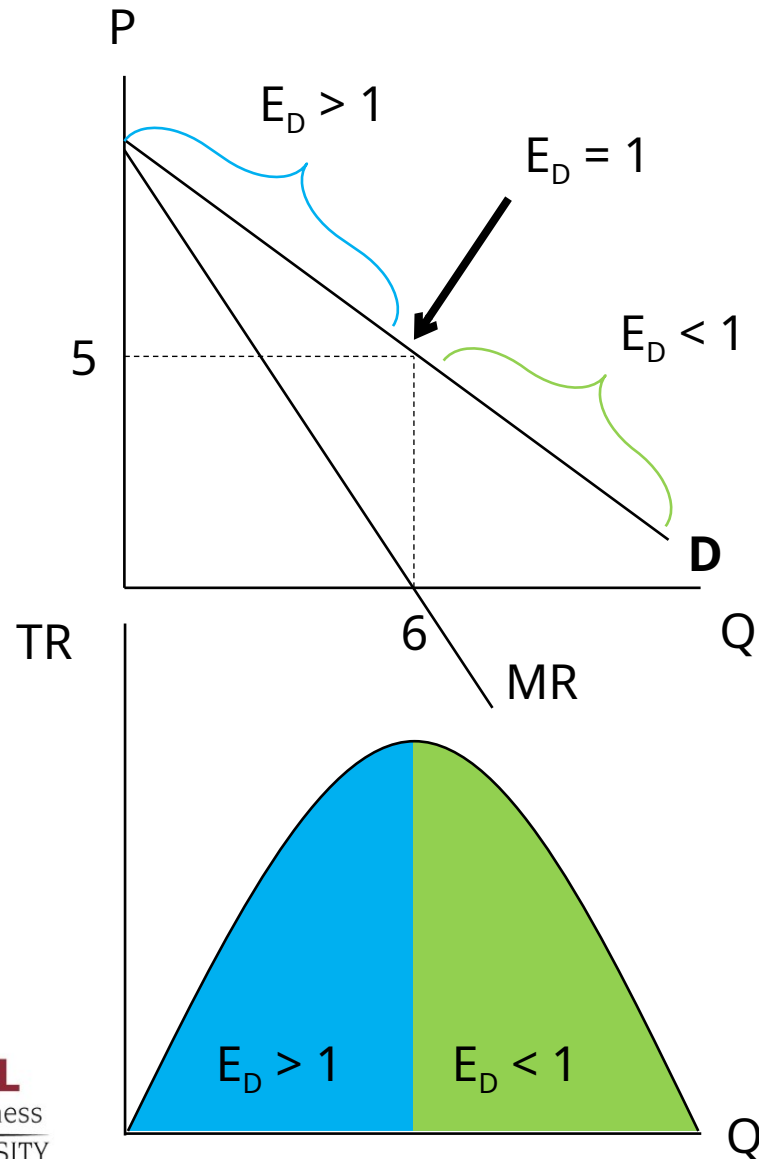
Price Elasticity and Total Revenue

P	Q	%ΔP	%ΔQ	TR	MR	Ed
10	1	-10%	100%	10	10	21.0
9	2	-11.1%	50%	18	8	6.33
8	3	-12.5%	33.3%	24	6	3.40
7	4	-14.3%	25%	28	4	2.14
6	5			30	2	1.44
5	6			30	0	1.00
4	7			28	-2	0.69
3	8			24	-4	0.47
2	9			18	-6	0.29
1	10			10	-8	0.16



$$\% \Delta \text{Revenue} = \% \Delta \text{Price} + \% \Delta \text{Quantity}$$

Price Elasticity and Total Revenue



- Output effect is greater than price effect when elasticity is greater than 1.
- Price Effect is greater than output effect when elasticity is less than 1.
- Q1: If $E_D = 0.3$, should you raise or lower price?
 - Raise price
- Q2: A movie theater has an elasticity of demand = 2.1, should it raise or lower price?
 - Lower price

Activity – Maximizing Revenue

Scenario:

- You priced your product at \$10 and sold 80 units. You tried a price of \$15 and sold 70 units.

Questions

1. What is your arc price elasticity of demand?
2. Did you make the right decision by increasing price from \$10 to \$15?

Activity – Maximizing Revenue

Step 1: Solve for arc price elasticity:

$$E_D = \frac{Q_2 - Q_1}{P_2 - P_1} * \frac{P_2 + P_1}{Q_2 + Q_1}$$

$$E_D = \frac{70 - 80}{15 - 10} * \frac{15 + 10}{70 + 80}$$

$$E_D = \frac{-10}{5} * \frac{25}{150}$$

$$E_D = -1/3$$

Step 2: Determine if you made the right decision.

A price elasticity of -0.33 implies you have inelastic demand. Raising price to \$15 was the correct move as it raises total revenue (\$1,050 at \$15 compared to \$800 at \$10).

You need to continue to raise your price to maximize revenue.

Total revenue will be maximized when the price elasticity of demand = 1.

Price Elasticity and Optimal Pricing Policy

- Optimal Price Formula
 - MR and ε_p are directly related.
 - $MR = P \times [1 + (1/\varepsilon_p)]$.
 - Optimal $P^* = MC/[1 + (1/\varepsilon_p)]$.

You have the following demand function with a current quantity of 900 and price of \$22

$$Q = 2000 - 50P$$

What is the optimal price if marginal costs = \$5?

Answer

$$Q = 900, P = \$22$$

$$\varepsilon_D = -50 * \frac{22}{900} = -1.22$$

$$P = \frac{MC}{1 + 1/E_D}$$

$$P = \frac{5}{1 + 1/-1.22}$$

$$P = \$27.74$$

$$MR = P \times [1 + (1/\varepsilon_p)]$$

$$MR = 27.74 \times [1 + (1/-1.22)]$$

$$MR = 27.74 \times [1 + (1/-1.22)] = \$5$$

If current price is not at \$27.74, consider adjusting it to maximize profit.

What Decision to Make?

- Recall that we want to produce the profit maximizing level of output.
 - This occurs where $MR = MC$
- This lets us revert to what we learned in previous lectures
 - 1) If $MR > MC$ then lower price and sell more
 - 2) If $MC > MR$ then raise price and sell less
- When making these decisions be careful not to confuse the average revenue with marginal revenue.
 - Remember the marginal revenue curve slopes down faster than the average revenue curve (the demand curve)
 - This lets us know that an additional unit sold will lower marginal revenue (the additional gain in revenue) faster than total revenue.

Determinants of Elasticity

- Close substitute = more elastic demand
 - iPhone and Samsung Galaxy
- Individual demand is more elastic than industry demand
 - Budweiser, Beer, Alcohol
- Products with many complements = less elastic demand
 - Kindle and eBooks
- Amount of budget spent
 - Candy bar vs. house
- Time horizon
 - Consider gasoline prices at different time intervals and individual demand

Case Study: Walmart vs. Amazon: Profit vs. Growth

- Walmart entered the online retail market in the early 2000s but may have faced a dip in its margin there as it continued to focus on brick-and-mortar and in-store pickup, as well as, Amazon having extensive capital investment.
 - By 2016, Walmart acquired Jet.com to bolster its online retail.
- In the early 2010s, Amazon had negative net income or near zero income with negative or near zero profit margins. Walmart maintained positive net income with margins around 3.5% to 4%.
- What was going on?
 - Amazon was priming for growth by focusing on sales volume, while Walmart has been trying to achieve higher profits.
 - What does this mean? That Amazon is likely not pricing where $MR = MC$, whereas Walmart was.
 - A key strategy for Amazon could have been pricing products to be loss-leaders (i.e. selling products at a low price to attract consumer in hopes they may buy more profitable items as well.
 - Amazon faced higher price elasticities in items like books, electronics and apparel, while Walmart faced lower ones in items like groceries and essentials.
 - Sources: [Managerial Econ Blog](#), [Financial Times](#)

Discussion Activity 3

- Why did Amazon focus on volume and growth while Walmart focused on margin and profit?
- Consider elasticity differences, online vs. in-store strategies, and long-run goals.

Cross-price Elasticity of Demand

- Cross-price elasticity shows demand sensitivity to changes in other prices.
 - $\epsilon_{PX} = \partial Q_Y / Q_Y \div \partial P_X / P_X$.
- Substitutes have $\epsilon_{PX} > 0$.
 - E.g., *Coke* demand and *Pepsi* prices.
- Complements have $\epsilon_{PX} < 0$.
 - E.g., *Coke* demand and *Fritos* prices.
- Independent goods have $\epsilon_{PX} = 0$.
 - E.g., *Coke* demand and car prices.
- It's important to not take the absolute value of this elasticity as the sign determines what type of good we are dealing with: substitute or complement

Income Elasticity of Demand

- Income elasticity shows demand sensitivity to changes in income.
 - $\epsilon_I = \partial Q/Q \div \partial I/I$.
- Normal goods have $\epsilon_I > 0$.
 - A 5% decrease in income leads to a 10% decrease in quantity demanded
 - $-10\%/-5\% = 2$
- Inferior goods have $\epsilon_I < 0$.
 - A 5% increase in income leads to a 10% decrease in quantity demanded
 - $-10\%/5\% = -2$
- It's important to not take the absolute value of this elasticity as the sign determines what type of good we are dealing with: normal or inferior

Interpreting Other Elasticities

Type	Elasticity	Factor Change	Predicted Δ Quantity
Income	1.2	+5% income	+6%
Cross-price	-0.8	-10% price of complement	+8%
Cross-price	0.5	+20% price of substitute	+10%

Stay-Even Analysis

- Stay-even analysis lets you determine when a price increase become unprofitable, *i.e.* how much output you can afford to lose and still remain at the same profit level.

$$\% \Delta Q = \frac{\% \Delta P}{\% \Delta P + \text{margin}}$$

- where $\text{margin} = (P - MC)/P$
- A bakery currently sells chocolate chip cookies at a price of \$16 per dozen. The marginal cost per dozen is \$8. The cookies are becoming more popular with customers and so the bakery owner is considering raising the price to \$20/dozen. What percentage of customers can be lost to ensure that the price increase is profitable?

$$\text{margin} = \frac{\$20 - \$8}{\$20} = 60\% \quad \text{price change \%} = \frac{\$20 - \$16}{\$16} = 25\%$$

$$\% \Delta Q = \frac{25\%}{25\% + 60\%} = 29.41\%$$

Activity – Should You Raise the Price?

Scenario:

- You run a bakery selling cupcakes at \$2.00 each. You're considering raising the price to \$2.25. Each cupcake costs \$1.25 to make. You currently sell 1,000 cupcakes/week.

Questions:

1. How many customers can you afford to lose and still make the same profit?
2. Follow-up: Would your answer change if this were a premium cupcake with few substitutes?

Activity – Should You Raise the Price?

Given:

- Current price: \$2.00
- New price: \$2.25
- Cost per cupcake: \$1.25
- Current sales: 1,000 units/week

Calculate Contribution Margins:

- Old Margin = $\$2.00 - \$1.25 = \$0.75$
- New Margin = $\$2.25 - \$1.25 = \$1.00$

Stay-Even Quantity:

Stay-even quantity = $(0.75 / 1.00) \times 1,000 = 750$ cupcakes

- You can afford to lose up to 250 customers (25%) and still earn the same total contribution.

Cost-Based Pricing

- Cost-based pricing is also known as cost plus pricing

$$m = \frac{P - C}{C}$$

m is the markup, P is price, and C is average (variable) cost

$$P = C(1 + m)$$

- Firms may not often know their demand, so they charge a percentage over the cost of the item.
- Why does it occur?
 - Because the people making the pricing decisions do not have enough information.
 - They know their costs, but don't know their demand
 - It is more than likely to be suboptimal
 - Caveat: it does allow stability in pricing if costs don't change

Cost-Based Pricing

- › We can approximate the optimal price for a single product by remembering that $MR = MC$.
- Start with the following:

$$MR = P \left(1 + \frac{1}{E} \right)$$

- Next, rearrange to isolate P:

$$P = MR \frac{E}{E + 1}$$

- Since we are trying to profit maximize ($MR = MC$), substitute MC for MR

$$P = MC \frac{E}{E + 1}$$

- Next, if the firm is maintaining a constant level of costs, replace MC with C:

$$P = C \frac{E}{E + 1}$$

- From the previous slide we know $P = C(1+m)$

$$C(1 + m) = C \frac{E}{E + 1}$$

- Rearrange to get:

$$1 + m = \frac{E}{E+1} \quad \rightarrow \quad m = \frac{E}{E+1} - 1$$

Key Takeaways

- Pricing is an extent decision: Increase price if $MR < MC$; lower price if $MR > MC$.
- Demand elasticity determines how price affects quantity and revenue.
- Elastic demand ($|E| > 1$) → quantity drops more than price rises
Inelastic demand ($|E| < 1$) → price rises more than quantity drops
- Point vs. arc elasticity: use arc when data is discrete, point when using calculus or functions.
- Elasticity depends on:
 - Substitutes
 - Budget share
 - Time frame
 - Product type (necessity/luxury)
- Use cross-price elasticity to assess relationships between goods.
→ Substitutes = positive; complements = negative
- Stay-even analysis helps firms assess if they can profitably change prices.
- Strategic pricing depends on elasticity, margin, and long-run business goals.

Knowledge Check

1. What is the difference between individual and aggregate demand?
2. When should a firm raise price vs. lower it, using marginal analysis?
3. What does it mean if the price elasticity of demand is greater than 1?
4. What is cross-price elasticity and how is it interpreted?
5. How can stay-even analysis help in pricing decisions?

Knowledge Check Answers

1. Individual = one consumer's demand; Aggregate = total demand across all consumers.
2. Raise price if $MR < MC$; lower price if $MR > MC$.
3. Demand is elastic — quantity responds strongly to price changes.
4. Measures how demand for one good responds to price change in another.
 - Positive = substitutes; Negative = complements
5. Helps determine how much quantity can fall after a price increase before profit declines.

Consultant's Challenge: Pricing for Profit

Scenario:

- You're advising a local meal-prep business that recently raised its prices and immediately saw a drop in orders. They aren't sure whether to hold firm, cut prices back, or invest in more advertising.

Your Task:

1. How would you help them estimate elasticity and interpret what it says about their customers' sensitivity to price?
2. What pricing strategy or adjustment would you recommend and how would you justify it using demand data?